AI-Based Smart Electric Vehicle Charging: A Review of Load Balancing, Reservation, Reliability and Waiting-Time Prediction

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***Abstract*—Electric vehicle (EV) adoption is increasing the demand for reliable and efficient charging infrastructure. Recent research has addressed EV charging through multi-objective station allocation, smart-grid energy management, AI-based demand prediction, dynamic load balancing, reservation, waiting-time forecasting, occupancy prediction, and charging-infrastructure reliability analysis. This paper reviews these approaches and compares their methodologies, evaluation measures, reported outcomes, and identified limitations. The reviewed studies show that optimization methods are effective for grid planning, AI methods support demand forecasting and adaptive charging, reservation systems can balance charging requests, and waiting-time or occupancy prediction can improve charging decisions. Reliability-oriented methods further indicate that user-generated information can help identify charging-station problems. However, the reviewed approaches generally address these issues separately. A major research gap is the absence of a unified, user-centric framework that combines predictive availability, waiting-time estimation, station reliability, charging cost, route suitability, and reservation. This review therefore motivates an integrated AI-enabled EV charging system that can improve both infrastructure utilization and the user charging experience.**

***Index Terms*—electric vehicles; smart charging; machine learning; load balancing; reservation; charging-station reliability; waiting-time prediction; occupancy prediction**

I. Introduction

Electric vehicles are becoming an important part of sustainable transportation. However, the increasing number of EVs also creates operational challenges for public charging infrastructure. Charging demand varies significantly with time, location, vehicle characteristics, and user behaviour. Consequently, some charging stations may become congested while nearby stations remain underutilized. Long waiting times, uneven station utilization, high charging cost, and additional stress on the electrical grid can reduce the overall charging experience.

The reviewed literature demonstrates that these challenges have been investigated from different perspectives. Grid-oriented studies concentrate on power loss, voltage profile, energy management, renewable-energy integration, and vehicle-to-grid (V2G) operation [1], [2]. AI-based approaches use machine learning, predictive analytics, demand response, dynamic pricing, and load balancing to adapt charging to changing demand [3], [4]. Other studies focus on reservation and route planning [5], while recent work predicts waiting time and charging-station occupancy [6], [7]. Reliability can also be estimated from charging-station reviews using deep-learning methods [8].

The objective of this paper is to organize these approaches into a common view and identify the limitations that should be addressed by a more integrated EV charging system.

II. EV Charging and Grid-Oriented Approaches

Multi-objective charging-station allocation methods consider several electrical and planning objectives simultaneously. The reviewed 2023 work evaluates power loss, voltage profile, operational cost, and renewable-energy utilization using the IEEE 69-bus system [1]. Such methods are useful when deciding where charging stations and renewable sources should be placed, but they do not directly provide real-time slot booking, predictive availability, or user-oriented charging services.

Smart-grid integration has also been investigated through V2G and grid-to-vehicle (G2V) scheduling. The Improved Honey Badger Algorithm (IHBA) is used to optimize EV energy management and evaluate efficiency, power loss, and total harmonic distortion [2]. These methods demonstrate the value of intelligent charging and discharging, but reservation, navigation, real-time charger availability, and prepaid booking are outside their main scope.

Standards-based communication is another important component of smart charging. A public charging infrastructure study uses ISO 15118-based communication and demonstrates smart charging at a real charging hub [3]. The approach supports interoperable charging control, but it does not provide reservation, route planning, prepaid booking, or charger-reliability prediction.

III. AI-Based Smart Charging and Load Balancing

AI-based charging systems attempt to predict demand and adapt charging decisions to the expected load. The reviewed Malaysian study combines machine learning, predictive analytics, demand response, dynamic pricing, V2G integration, and mathematical optimization. It reports 30% energy savings and 20.38% cost reduction in the reviewed material [4].

Another 2025 approach combines predictive demand forecasting, machine learning and deep learning, dynamic load balancing, demand response, smart-grid integration, secure

energy management, and V2G [9]. The review reports a 20% reduction in peak-period overload. These results indicate that AI can contribute to both grid stability and charging-cost reduction.

Despite these advantages, the reviewed AI approaches have limitations. The Malaysian work identifies limited real-world deployment, insufficient localized datasets, and difficulty with large-scale implementation [4]. The load-balancing study also identifies scalability and centralized-system vulnerabilities [9]. Therefore, AI-based load balancing should be complemented by mechanisms that consider individual station availability and user requirements.

IV. Reservation and Charging-Station Utilization

Reservation-based charging directly addresses the mismatch between charging demand and station capacity. The reservation-based approach reviewed in [5] allows EV users to book a charging station before arriving and evaluates the method using real charging-station data. The reported benefits include reduced waiting time, improved station usage, and better distribution of charging requests among stations.

Route-oriented reservation research uses the K-Shortest Path algorithm to select charging stations by considering traffic, distance, charging cost, and station availability. The reviewed study reports reductions in travel time and charging cost [10]. However, it does not use AI or machine learning and concentrates mainly on route planning and reservation.

The Business Intelligence model reviewed in [11] uses AI and machine learning to analyze EV and charging-station data and recommend charging schedules. It reports reduced waiting time, improved station usage, reduced charging cost, and improved schedule recommendations. Its main limitations are the need for large quantities of real-time data, secure data sharing, and scalability.

V. Waiting-Time and Occupancy Prediction

Waiting-time prediction is particularly important for EV users because the charging decision is affected not only by distance and price but also by the expected time spent at a station. A 2025 study proposes Mixed-Integer Dynamic Programming (MIDP) integrated with Monte Carlo simulation to forecast waiting times at fast charging stations and optimize long-distance charging plans [6]. The reviewed results indicate up to 9.6% reduction in total trip time and up to 15.1% reduction in charging waiting time.

The same work assumes fixed traffic and accurate forecasts and does not fully incorporate real-time traffic, dynamic pricing, renewable-energy integration, or large-scale validation [6]. These limitations indicate the need for continuously updated prediction models.

Adaptive federated learning has been proposed for EV charging-station occupancy prediction [7]. The method uses MAE, RMSE, and MAPE and improves forecasting while preserving data privacy. However, further evaluation on larger charging networks and diverse real-time traffic conditions is required. Privacy-preserving prediction is nevertheless promising for networks operated by multiple charging providers.

Fig. 1 illustrates the large variation in energy consumed across EV charging instances. This variability supports the use of demand-aware scheduling rather than treating every charging event as identical.

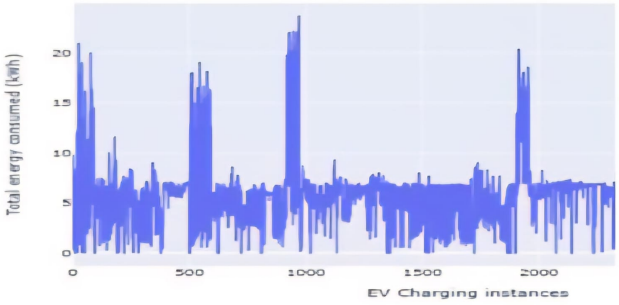


Fig. 1. Energy consumed by EV charging instances [9].

VI. Charging Infrastructure Reliability

Availability alone does not guarantee a satisfactory charging experience. A station may be shown as available while having operational problems, damaged connectors, or other service issues. The reviewed cross-lingual deep-learning study analyzes charging-station reviews using BERT-based deep learning and machine translation [8]. It reports 89.1% accuracy and an F1-score of 0.78 for identifying charging-station problems.

The approach demonstrates that user-generated information can complement conventional station data. However, it depends on translated user reviews and requires stronger support for real-time multilingual information and broader real-world testing [8]. This creates an opportunity to combine review-based reliability with sensor-based availability and historical station performance.

VII. Comparative Discussion

The reviewed research demonstrates progress in individual aspects of EV charging. Optimization-based methods are strong in grid planning and electrical constraints [1], [2]. AI-based methods are suitable for demand forecasting, dynamic load balancing, and adaptive charging [4], [9]. Reservation approaches improve request distribution and station utilization [5], [10], [11]. Waiting-time and occupancy prediction provide information that can improve charging decisions before a vehicle reaches a station [6], [7]. Reliability analysis adds another user-oriented dimension [8].

These methods are complementary rather than competing. A practical charging system should therefore use a layered architecture in which prediction models estimate future demand and occupancy, a reliability module evaluates station dependability, and a decision module ranks stations according to availability, waiting time, cost, distance, and reliability. A reservation component can then secure a selected charging slot.

TABLE I
COMPARATIVE SUMMARY OF REVIEWED STUDIES

#	Technique Used	Outcomes	Research Gap
1	Multi-objective charging-station allocation [1]	Evaluates power loss, voltage profile, operational cost, and renewable-energy utilization.	No real-time slot booking, predictive availability, or user-oriented charging services.
2	V2G/G2V energy management [2]	IHBA optimizes EV energy management, efficiency, power loss, and total harmonic distortion.	Reservation, navigation, real-time availability, and prepaid booking are outside scope.
3	ISO 15118 smart charging communication [3]	Demonstrates interoperable smart charging control at a real charging hub.	No reservation, route planning, prepaid booking, or charger-reliability prediction.
4	AI smart charging in Malaysia [4]	Reports 30% energy savings and 20.38% cost reduction.	Limited real-world deployment, localized datasets, and large-scale implementation.
5	Reservation-based charging [5]	Reduces waiting time, improves station usage, and distributes charging requests.	Does not integrate the full set of predictive and reliability factors.
6	Waiting-time forecast with MIDP and Monte Carlo [6]	Up to 9.6% lower total trip time and 15.1% lower charging waiting time.	Fixed traffic assumptions, forecast dependence, and limited large-scale validation.
7	Federated occupancy prediction [7]	Uses MAE, RMSE, and MAPE while preserving data privacy.	Needs evaluation on larger networks and diverse real-time traffic conditions.
8	Cross-lingual deep-learning reliability analysis [8]	Reports 89.1% accuracy and 0.78 F1-score for identifying station problems.	Depends on translated reviews; needs stronger real-time multilingual testing.
9	AI load balancing and grid integration [9]	Combines forecasting, ML/DL, dynamic load balancing, demand response, and V2G; reports 20% lower peak overload.	Scalability and centralized-system vulnerabilities.
10	K-Shortest Path route reservation [10]	Considers traffic, distance, cost, and station availability; reports lower travel time and charging cost.	No AI/ML; mainly focused on route planning and reservation.
11	AI/ML Business Intelligence charging schedules [11]	Reports reduced waiting time and charging cost with improved schedule recommendations.	Requires large real-time datasets, secure data sharing, and scalability.

Fig. 2 presents an AI-based smart charging architecture in which data are pre-processed, an AI model performs prediction, and dynamic load balancing, grid optimization, and V2G are used before real-time tracking and evaluation. This figure summarizes the infrastructure-side intelligence identified across several reviewed studies.

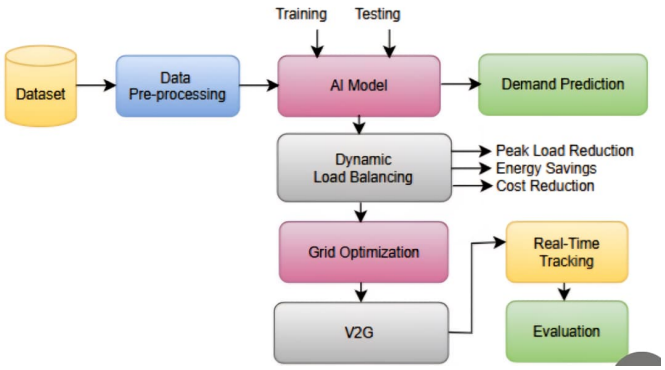


Fig. 2. Block diagram of an AI-based method for smart EV charging using load balancing [4], [9].

The main challenge is obtaining sufficiently accurate and timely data while maintaining scalability, privacy, and secure communication. Federated learning and standards-based charging communication provide possible solutions for these requirements.

Fig. 3 compares average station utilization before and after simulation. The difference between the two cases indicates that intelligent charging management can redistribute utilization across stations.

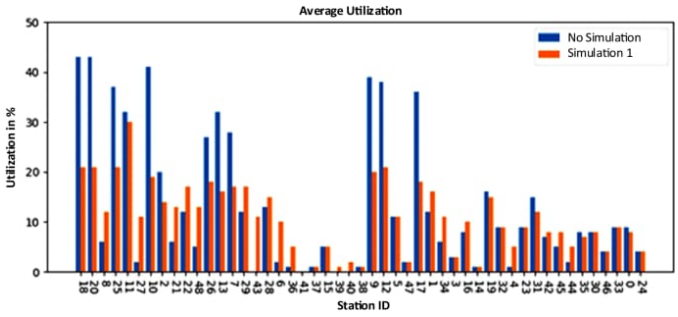


Fig. 3. Average utilization of charging stations before and after simulation [5].

VIII. RESEARCH GAP AND PROPOSED DIRECTION

The major gap identified from the reviewed papers is fragmentation. Grid optimization generally focuses on infrastructure performance, while reservation methods focus on request distribution. AI methods improve prediction and load balancing, but may not incorporate station reliability. Waiting-time forecasting improves trip planning, but often relies on assumptions about traffic and forecast accuracy. Reliability analysis provides valuable user information but is not normally integrated with charging reservation and route selection.

Fig. 4 represents charging-station usage and the distribution of charging cost. Together, these plots demonstrate why station selection should consider both utilization and cost.

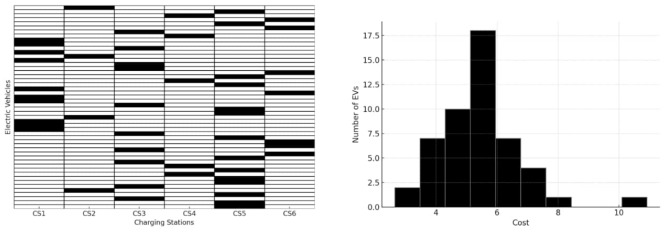


Fig. 4. Charging-station usage and distribution of charging cost [11].

Fig. 5 presents a framework for forecasting vehicle arrival and waiting time at a fast charging station. It uses vehicle arrival information, charging-port availability, battery characteristics, BEV models, and arrival state-of-charge information to estimate waiting time.

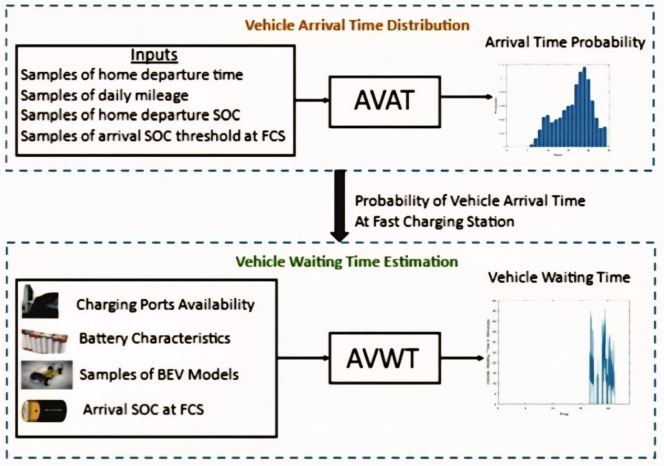


Fig. 5. Framework for forecasting vehicle waiting time at a fast charging station [6].

Based on these observations, an integrated AI-enabled EV charging framework can be organized into four stages. First, charging records, station occupancy, cost, location, user reviews, and vehicle requirements are collected and preprocessed. Second, predictive models estimate demand, occupancy, and waiting time, while a reliability model produces a station reliability score. Third, a decision module ranks candidate stations using distance, expected waiting time, cost, predicted availability, and reliability. Finally, a reservation module secures the selected charging slot and updates the decision when new real-time information becomes available.

This direction combines the strongest ideas identified in the reviewed literature and directly addresses the missing combination of predictive availability, reliability, route-aware decision making, and reservation.

IX. Conclusion

This paper reviewed recent research on EV charging station allocation, smart-grid energy management, AI-based load balancing, reservation, charging-station utilization, waiting-time prediction, occupancy forecasting, and infrastructure reliability. The reviewed studies show that intelligent methods can reduce energy consumption, cost, peak overload, waiting time, and inefficient station utilization. However, most existing approaches address only one or two of these objectives.

The review therefore identifies a need for an integrated, user-centric charging framework that combines prediction, reliability assessment, cost and distance awareness, waiting-time estimation, and reservation. Future work should implement and evaluate such a framework using real-time charging-station data and realistic EV charging scenarios.

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